Associate Professor in Data Science
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1 Highlights

Researcher in Data Science (modelling of complex phenomena and developing *usable*, general-purpose algorithms and data analysis software)

- Research interests: machine learning, artificial intelligence, data clustering, data fusion, aggregation, *metrics, prototype learning, computational statistics, mathematical modelling for sports analytics, informetrics, economics, science of science, etc.
- Area editor (aggregation functions and data science) in *Fuzzy Sets and Systems*
- Author/editor of 101 publications, including journal papers in outlets such as *Proceedings of the National Academy of Sciences (PNAS)*, *Journal of Statistical Software*, *R Journal*, *Journal of Classification*, *Information Fusion*, *International Journal of Forecasting*, *Statistical Modelling*, *Physica A: Statistical Mechanics and Its Applications*, *Information Sciences*, *Knowledge-Based Systems*, *IEEE Transactions on Fuzzy Systems*, and *Journal of Informetrics*
- Current h-index = 23 (G**gle Scholar)

Free (Libre) and Open Source Data Analysis Software Developer

- Author and maintainer of *stringi* – one of the most often downloaded R packages (text/natural language processing; over 100,000,000 downloads)
- Author and maintainer of the fast and robust *Genie* hierarchical clustering algorithm (see the Python and R package *genieclust*)

Data Science, Machine Learning, and Statistical Computing Lecturer/Tutor

- Current: Warsaw University of Technology (Warsaw, Poland)
- Past: Deakin University (Melbourne, Australia), Data Science Retreat (Berlin, Germany)
- Author of the open-access textbooks *Deep R Programming* and *Minimalist Data Wrangling with Python*

2 Qualifications

11.2020	Deakin University, Melbourne, VIC, Australia *Graduate Certificate* of Higher Education: Learning And Teaching (E575)
10.2017	Systems Research Institute, Polish Academy of Sciences, Warsaw, Poland *DSc* (*Habilitation* – Higher Doctorate) in Computer Science
12.2011	Systems Research Institute, Polish Academy of Sciences, Warsaw, Poland *PhD* in Computer Science
06.2008	Faculty of Mathematics and Information Science, Warsaw University of Technology, Poland *BEng and MSc* (by Research) in Computer Science (cum laude) [GPA 4.964/5.000]

3 Employment History

04.2024 –	Faculty of Mathematics and Information Science, Warsaw University of Technology, Poland Department of Computational Statistics and Data Analysis *Associate Professor in Data Science* (05.2024 –) *Academic Director for Data Science studies* (09.2025 –) *Vice-chair of the Teaching and Learning Committee for BSc and MSc in Data Science* (04.2024 –)
04.2024 –	Department of Stochastic Methods, Systems Research Institute, Polish Academy of Sciences, Warsaw, Poland Department of Stochastic Methods *Associate Professor* (04.2024 –)
09.2019 – 03.2024	School of Information Technology, Deakin University, Melbourne, VIC, Australia *Senior Lecturer in Applied Artificial Intelligence/Data Science* [50% research/40% teaching/10% service load; corresponds to an Associate Professor position in North American universities (tenured)] (09.2019 – 03.2024) *Deputy Course Director – BSc Degree in Data Science* (03.2022 – 03.2024)
10.2008 – 09.2019	Faculty of Mathematics and Information Science, Warsaw University of Technology, Poland *Associate Professor in Data Science* (01.2018 – 09.2019) *Academic Director for Data Science studies* (01.2018 – 09.2019) *(Founding) Vice-chair of the Teaching and Learning Committee for BSc and MSc in Data Science* (10.2016 – 09.2019) *Assistant Professor* (04.2012 – 12.2017) *Teaching and Research Assistant* (10.2008 – 02.2012)
07.2008 – 08.2019	Department of Stochastic Methods, Systems Research Institute, Polish Academy of Sciences, Warsaw, Poland *Associate Professor* (04.2018 – 08.2019) *Assistant Professor* (02.2012 – 03.2018) *Research Assistant* (07.2008 – 01.2012)

4 Short-term Research Visits and Casual Employment

07.2017 – 08.2017	Deakin University, Melbourne, VIC, Australia School of Information Technology *Visiting Professor*
04.2015 – 06.2015	IRAFM, University of Ostrava, Czechia *Visiting Professor*
07.2014 – 07.2019	Data Science Retreat, Berlin, Germany *Python, R, and Data Science Tutor* (19 batches)
03.2013 – 06.2013	Slovak University of Technology in Bratislava, Slovakia *Visiting Professor*

5 Scholarships and Awards

1. Included in the Stanford/Elsevier Top 2% Scientists List for both 2024 and 2025
2. Warsaw University of Technology award for excellence in research, 2026 (individual, 1st degree)
3. The President of Poland, the Bronze Medal for Long Service, 2025
4. Ministry of Education and Science, Poland, award for significant achievements in teaching: design and implementation of a new innovative course of study – Master of Data Science – at the Faculty of Mathematics and Information Science, Warsaw University of Technology, 2022 (team)
5. Deakin University's School of IT best paper award, 2022
6. Warsaw University of Technology award for excellence in research, 2024 (with A. Cena, 1st degree)
7. Warsaw University of Technology award for excellence in research, 2022 (with M. Bartoszuk and A. Cena, 1st degree)
8. Warsaw University of Technology IDUB PW best paper award (3 papers), 2020
9. Warsaw University of Technology award for excellence in research, 2020 (individual, 1st degree)
10. Golden Chalk for teaching excellence, Faculty of Mathematics and Information Science, Warsaw University of Technology, 2019 (individual, 2nd degree, i.e., silver)
11. Warsaw University of Technology award for excellence in teaching, 2017 (with M. Bartoszuk and A. Cena, 3rd degree)
12. Ministry of Science and Higher Education, Poland, scholarship for young researchers, 2015 (36 months)
13. Warsaw University of Technology award for excellence in teaching, 2015 (with K. Bobecka-Wesołowska and P. Grzegorzewski, 3rd degree)
14. Foundation for Polish Science (FNP), scholarship for young researchers – START Program, 2013 (12 months)
15. Warsaw University of Technology award for excellence in research, 2012 (with P. Grzegorzewski, 1st degree)
16. Warsaw University of Technology award for excellence in research, 2010 (with P. Grzegorzewski, 1st degree)
17. Ministry of Science and Higher Education, Poland, scholarship for research achievements for students, 2007

6 Research Activities

6.1 Research Projects

1. Chief Investigator (02.2021–03.2024), Partner Investigator (04.2024–02.2025): Australian Research Council, Discovery Project ARC DP210100227, *Beyond black-box models: Interaction in eXplainable Artificial Intelligence*, Deakin University, Australia; Lead CI: Gleb Beliakov; 2021 (48 months)
2. Partner Investigator: The Czech Science Foundation, research project GAČR 18-06915S, *New approaches to aggregation operators in analysis and processing of data*, University of Olomouc, Czechia; Lead CI: Radomír Halaš; 2018 (36 months)

3. Lead Chief Investigator: National Science Centre, Poland, research project NCN Sonata 2014/13/D/HS4/01700, *Construction and analysis of methods of information resources producers' quality management*, Systems Research Institute, Polish Academy of Sciences; 2015 (30 months)

6.2 Publications

6.2.1 Journal Articles (56)

P1. Brzozowski, Ł., Gagolewski, M., Siudem, G., Generating synthetic citation networks with communities, *Information Sciences* **761**, 124196, 2027

P2. Brzozowski, Ł., Gagolewski, M., Siudem, G., Żogała-Siudem, B., The Price–Pareto growth model of networks with community structure, *Journal of Informetrics* **20**(3), 101850, 2026

P3. Gagolewski, M., Normalised clustering accuracy: An asymmetric external cluster validity measure, *Journal of Classification* **42**, 2–30, 2025

P4. Gagolewski, M., Cena, A., Bartoszuk, M., Brzozowski, Ł., Clustering with minimum spanning trees: How good can it be?, *Journal of Classification* **42**, 90–112, 2025

P5. Bertoli-Barsotti, L., Gagolewski, M., Siudem, G., Żogała-Siudem, B., Equivalence of inequality indices in the three-dimensional model of informetric impact, *Journal of Informetrics* **18**(4), 101566, 2024

P6. Bertoli-Barsotti, L., Gagolewski, M., Siudem, G., Żogała-Siudem, B., Gini-stable Lorenz curves and their relation to the generalised Pareto distribution, *Journal of Informetrics* **18**(2), 101499, 2024

P7. Wu, J-Z., Beliakov, G., James, S., Gagolewski, M., Random generation of linearly constrained fuzzy measures and domain coverage performance evaluation, *Information Sciences* **659**, 120080, 2024

P8. Gagolewski, M., Cena, A., James, S., Beliakov, G., Hierarchical clustering with OWA-based linkages, the Lance–Williams formula, and dendrogram inversions, *Fuzzy Sets and Systems* **473**, 108740, 2023

P9. Boczek, M., Gagolewski, M., Kaluszka, M., Okolewski, A., A benchmark-type generalization of the Sugeno integral with applications in bibliometrics, *Fuzzy Sets and Systems* **466**, 108479, 2023

P10. Żogała-Siudem, B., Cena, A., Siudem, G., Gagolewski, M., Interpretable reparameterisations of citation models, *Journal of Informetrics* **17**(1), 101355, 2023

P11. Gagolewski, M., A framework for benchmarking clustering algorithms, *SoftwareX* **20**, 101270, 2022, URL: https://clustering-benchmarks.gagolewski.com/

P12. Siudem, G., Nowak, P., Gagolewski, M., Power laws, the Price Model, and the Pareto type-2 distribution, *Physica A: Statistical Mechanics and its Applications* **606**, 128059, 2022

P13. Gagolewski, M., stringi: Fast and portable character string processing in R, *Journal of Statistical Software* **103**(2), 1–59, 2022, URL: https://stringi.gagolewski.com/

P14. Beliakov, G., Gagolewski, M., James, S., Reduction of variables and constraints in fitting antibuoyant fuzzy measures to data using linear programming, *Fuzzy Sets and Systems* **451**, 266–284, 2022

P15. Geras, A., Siudem, G., Gagolewski, M., Time to vote: Temporal clustering of user activity on Stack Overflow, *Journal of the Association for Information Science and Technology* **73**(12), 1681–1691, 2022

P16. Gagolewski, M., Żogała-Siudem, B., Siudem, G., Cena, A., Ockham's index of citation impact, *Scientometrics* **127**, 2829–2845, 2022

P17. Mrowiński, M., Gagolewski, M., Siudem, G., Accidentality in journal citation patterns, *Journal of Informetrics* **16**(4), 101341, 2022

P18. Cena, A., Gagolewski, M., Siudem, G., Żogała-Siudem, B., Validating citation models by proxy indices, *Journal of Informetrics* **16**(2), 101267, 2022

P19. Gagolewski, M., Bartoszuk, M., Cena, A., Are cluster validity measures (in)valid?, *Information Sciences* **581**, 620–636, 2021, URL: https://github.com/gagolews/optim_cvi

P20. Beliakov, G., Gagolewski, M., James, S., Hierarchical data fusion processes involving the Möbius representation of capacities, *Fuzzy Sets and Systems* **433**, 1–21, 2022

P21. Bartoszuk, M., Gagolewski, M., T-norms or t-conorms? How to aggregate similarity degrees for plagiarism detection, *Knowledge-Based Systems* **231**, 107427, 2021

P22. Lasek, J., Gagolewski, M., Interpretable sports team rating models based on the gradient descent algorithm, *International Journal of Forecasting* **37**(3), 1061–1071, 2021

P23. Gagolewski, M., genieclust: Fast and robust hierarchical clustering, *SoftwareX* **15**, 100722, 2021, URL: https://genieclust.gagolewski.com/

P24. Pérez-Fernández, R., Gagolewski, M., De Baets, B., On the aggregation of compositional data, *Information Fusion* **73**, 103–110, 2021

P25. Beliakov, G., Gagolewski, M., James, S., DC optimization for constructing discrete Sugeno integrals and learning nonadditive measures, *Optimization* **69**(12), 2515–2534, 2020

P26. Bartoszuk, M., Gagolewski, M., SimilaR: R Code Clone and Plagiarism Detection, *R Journal* **12**(1), 367–385, 2020, URL: https://CRAN.R-project.org/package=SimilaR

P27. Siudem, G., Żogała-Siudem, B., Cena, A., Gagolewski, M., Three dimensions of scientific impact, *Proceedings of the National Academy of Sciences of the United States of America (PNAS)* **117**, 13896–13900, 2020

P28. Coroianu, L., Fullér, R., Gagolewski, M., James, S., Constrained ordered weighted averaging aggregation with multiple comonotone constraints, *Fuzzy Sets and Systems* **395**, 21–39, 2020

P29. Cena, A., Gagolewski, M., Genie+OWA: Robustifying hierarchical clustering with OWA-based linkages, *Information Sciences* **520**, 324–336, 2020

P30. Gagolewski, M., Pérez-Fernández, R., De Baets, B., An inherent difficulty in the aggregation of multidimensional data, *IEEE Transactions on Fuzzy Systems* **28**, 602–606, 2020

P31. Beliakov, G., Gagolewski, M., James, S., Robust fitting for the Sugeno integral with respect to general fuzzy measures, *Information Sciences* **514**, 449–461, 2020

P32. Geras, A., Siudem, G., Gagolewski, M., Should we introduce a dislike button for academic papers?, *Journal of the Association for Information Science and Technology* **71**(2), 221–229, 2020

P33. Coroianu, L., Gagolewski, M., Grzegorzewski, P., Piecewise linear approximation of fuzzy numbers: Algorithms, arithmetic operations and stability of characteristics, *Soft Computing* **23**(19), 9491–9505, 2019, URL: https://CRAN.R-project.org/package=FuzzyNumbers

P34. Beliakov, G., Gagolewski, M., James, S., Aggregation on ordinal scales with the Sugeno integral for biomedical applications, *Information Sciences* **501**, 377–387, 2019

P35. Pérez-Fernández, R., De Baets, B., Gagolewski, M., A taxonomy of monotonicity properties for the aggregation of multidimensional data, *Information Fusion* **52**, 322–334, 2019

P36. Gagolewski, M., James, S., Beliakov, G., Supervised learning to aggregate data with the Sugeno integral, *IEEE Transactions on Fuzzy Systems* **27**(4), 810–815, 2019

P37. Beliakov, G., Gagolewski, M., James, S., Pace, S., Pastorello, N., Thilliez, E., Vasa, R., Measuring traffic congestion: An approach based on learning weighted inequality, spread and aggregation indices from comparison data, *Applied Soft Computing* **67**, 910–919, 2019

P38. Lasek, J., Gagolewski, M., The efficacy of league formats in ranking teams, *Statistical Modelling* **18**(5–6), 411–435, 2018

P39. Gagolewski, M., Penalty-based aggregation of multidimensional data, *Fuzzy Sets and Systems* **325**, 4–20, 2017

P40. Mesiar, R., Gagolewski, M., H-index and other Sugeno integrals: Some defects and their compensation, *IEEE Transactions on Fuzzy Systems* **24**(6), 1668–1672, 2016

P41. Beliakov, G., Gagolewski, M., James, S., Penalty-based and other representations of economic inequality, *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems* **24(Suppl.1)**, 1–23, 2016

P42. Gagolewski, M., Bartoszuk, M., Cena, A., Genie: A new, fast, and outlier-resistant hierarchical clustering algorithm, *Information Sciences* **363**, 8–23, 2016, URL: https://genieclust.gagolewski.com/

P43. Lasek, J., Szlavik, Z., Gagolewski, M., Bhulai, S., How to improve a team's position in the FIFA ranking – A simulation study, *Journal of Applied Statistics* **43**(7), 1349–1368, 2016

P44. Żogała-Siudem, B., Siudem, G., Cena, A., Gagolewski, M., Agent-based model for the bibliometric h-index – Exact solution, *European Physical Journal B* **89**(21), 2016

P45. Cena, A., Gagolewski, M., Mesiar, R., Problems and challenges of information resources producers' clustering, *Journal of Informetrics* **9**(2), 2015

P46. Cena, A., Gagolewski, M., OM3: Ordered maxitive, minitive, and modular aggregation operators – Axiomatic and probabilistic properties in an arity-monotonic setting, *Fuzzy Sets and Systems* **264**, 138–159, 2015

P47. Gagolewski, M., Spread measures and their relation to aggregation functions, *European Journal of Operational Research* **241**(2), 469–477, 2015

P48. Gagolewski, M., Mesiar, R., Monotone measures and universal integrals in a uniform framework for the scientific impact assessment problem, *Information Sciences* **263**, 166–174, 2014

P49. Gagolewski, M., On the relationship between symmetric maxitive, minitive, and modular aggregation operators, *Information Sciences* **221**, 170–180, 2013

P50. Coroianu, L., Gagolewski, M., Grzegorzewski, P., Nearest piecewise linear approximation of fuzzy numbers, *Fuzzy Sets and Systems* **233**, 26–51, 2013, URL: https://CRAN.R-project.org/package=FuzzyNumbers

P51. Gagolewski, M., Scientific impact assessment cannot be fair, *Journal of Informetrics* **7**(4), 792–802, 2013

P52. Gagolewski, M., Mesiar, R., Aggregating different paper quality measures with a generalized h-index, *Journal of Informetrics* **6**(4), 566–579, 2012

P53. Gagolewski, M., Bibliometric impact assessment with R and the CITAN package, *Journal of Informetrics* **5**(4), 678–692, 2011, URL: https://CRAN.R-project.org/package=CITAN

P54. Gagolewski, M., Grzegorzewski, P., Possibilistic analysis of arity-monotonic aggregation operators and its relation to bibliometric impact assessment of individuals, *International Journal of Approximate Reasoning* **52**(9), 1312–1324, 2011

P55. Gagolewski, M., Grzegorzewski, P., A geometric approach to the construction of scientific impact indices, *Scientometrics* **81**(3), 617–634, 2009

P56. Rowiński, T., Gagolewski, M., Preferencje i postawy wobec pomocy online (Attitudes towards online counselling and psychotherapy), *Studia Psychologica: Theoria et praxis* **7**, 195–210, 2007

6.2.2 Papers in Edited Volumes and Proceedings (35)

P57. Beliakov, G., James, S., Wu, J-Z., Gagolewski, M., *A Tractable Approach to Fitting the Choquet Integral for Explainable Prediction and Analysis*, in: *Proc. 2024 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, IEEE, 2024, pp. 10611753

P58. Coroianu, L., Gagolewski, M., *Penalty-based data aggregation in real normed vector spaces*, in: Halaš, R. et al. (eds.), *New Trends in Aggregation Theory*, Springer, 2019, pp. 160–171

P59. Beliakov, G., Gagolewski, M., James, S., *Least median of squares (LMS) and least trimmed squares (LTS) fitting for the weighted arithmetic mean*, in: Medina, J. et al. (eds.), *Information Processing and Management of Uncertainty in Knowledge-Based Systems. Theory and Foundations*, Springer, 2018, pp. 367–378

P60. Gagolewski, M., James, S., *Fitting symmetric fuzzy measures for discrete Sugeno integration*, in: Kacprzyk, J. et al. (eds.), *Advances in Fuzzy Logic and Technology 2017*, Springer, 2018, pp. 104–116

P61. Bartoszuk, M., Gagolewski, M., *Binary aggregation functions in software plagiarism detection*, in: *Proc. 2017 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, IEEE, 2017, pp. 8015582

P62. Cena, A., Gagolewski, M., *OWA-based linkage and the Genie correction for hierarchical clustering*, in: *Proc. 2017 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, IEEE, 2017, pp. 8015652

P63. Gagolewski, M., Cena, A., Bartoszuk, M., *Hierarchical clustering via penalty-based aggregation and the Genie approach*, in: Torra, V. et al. (eds.), *Modeling Decisions for Artificial Intelligence*, Springer, 2016, pp. 191–202

P64. Cena, A., Gagolewski, M., *Fuzzy k-minpen clustering and k-nearest-minpen classification procedures incorporating generic distance-based penalty minimizers*, in: Carvalho, J. et al. (eds.), *Information Processing and Management of Uncertainty in Knowledge-Based Systems, Part II*, Springer, 2016, pp. 445–456

P65. Bartoszuk, M., Beliakov, G., Gagolewski, M., James, S., *Fitting aggregation functions to data: Part I – Linearization and regularization*, in: Carvalho, J. et al. (eds.), *Information Processing and Management of Uncertainty in Knowledge-Based Systems, Part II*, Springer, 2016, pp. 767–779

P66. Bartoszuk, M., Beliakov, G., Gagolewski, M., James, S., *Fitting aggregation functions to data: Part II – Idempotization*, in: Carvalho, J. et al. (eds.), *Information Processing and Management of Uncertainty in Knowledge-Based Systems, Part II*, Springer, 2016, pp. 780–789

P67. Bartoszuk, M., Gagolewski, M., *Detecting similarity of R functions via a fusion of multiple heuristic methods*, in: Alonso, J., Bustince, H., Reformat, M. (eds.), *Proc. IFSA/EUSFLAT'15*, Atlantis Press, 2015, pp. 419–426

P68. Cena, A., Gagolewski, M., *A K-means-like algorithm for informetric data clustering*, in: Alonso, J., Bustince, H., Reformat, M. (eds.), *Proc. IFSA/EUSFLAT'15*, Atlantis Press, 2015, pp. 536–543

P69. Gagolewski, M., *Sugeno integral-based confidence intervals for the theoretical h-index*, in: Grzegorzewski, P. et al. (eds.), *Strengthening Links Between Data Analysis and Soft Computing*, Springer, 2015, pp. 233–240

P70. Gagolewski, M., *Normalized WD_pWAM and WD_pOWA spread measures*, in: Alonso, J., Bustince, H., Reformat, M. (eds.), *Proc. IFSA/EUSFLAT'15*, Atlantis Press, 2015, pp. 210–216

P71. Gagolewski, M., Lasek, J., *The use of fuzzy relations in the assessment of information resources producers' performance*, in: *Proc. 7th IEEE International Conference Intelligent Systems IS'2014, Vol. 2: Tools, Architectures, Systems, Applications*, Springer, 2015, pp. 289–300

P72. Gagolewski, M., Lasek, J., *Learning experts' preferences from informetric data*, in: Alonso, J., Bustince, H., Reformat, M. (eds.), *Proc. IFSA/EUSFLAT'15*, Atlantis Press, 2015, pp. 484–491

P73. Lasek, J., Gagolewski, M., *Estimation of tournament metrics for association football league formats*, in: *Selected problems in information technologies (Proc. ITRIA'15 vol. 2)*, Institute of Computer Science, Polish Academy of Sciences, 2015, pp. 67–78

P74. Lasek, J., Gagolewski, M., *The winning solution to the AAIA'15 Data Mining Competition: Tagging firefighter activities at a fire scene*, in: Ganzha, M., Maciaszek, L., Paprzycki, M. (eds.), *Proc. FedCSIS'15*, IEEE, 2015, pp. 375–380

P75. Cena, A., Gagolewski, M., *Clustering and aggregation of informetric data sets*, in: *Computational methods in data analysis (Proc. ITRIA'15 vol. 1)*, Institute of Computer Science, Polish Academy of Sciences, 2015, pp. 5–26

P76. Gagolewski, M., *Some issues in aggregation of multidimensional data*, in: Baczyński, M., De Baets, B., Mesiar, R. (eds.), *Proc. 8th International Summer School on Aggregation Operators (AGOP 2015)*, University of Silesia, 2015, pp. 127–132

P77. Cena, A., Gagolewski, M., *Aggregation and soft clustering of informetric data*, in: Baczyński, M., De Baets, B., Mesiar, R. (eds.), *Proc. 8th International Summer School on Aggregation Operators (AGOP 2015)*, University of Silesia, 2015, pp. 79–84

P78. Coroianu, L., Gagolewski, M., Grzegorzewski, P., Adabitabar Firozja, M., Houlari, T., *Piecewise linear approximation of fuzzy numbers preserving the support and core*, in: Laurent, A. et al. (eds.), *Information Processing and Management of Uncertainty in Knowledge-Based Systems, Part II*, Springer, 2014, pp. 244–254

P79. Bartoszuk, M., Gagolewski, M., *A fuzzy R code similarity detection algorithm*, in: Laurent, A. et al. (eds.), *Information Processing and Management of Uncertainty in Knowledge-Based Systems, Part III*, Springer, 2014, pp. 21–30

P80. Gagolewski, M., Dębski, M., Nowakiewicz, M., *Efficient algorithm for computing certain graph-based monotone integrals: The l_p-indices*, in: Mesiar, R., Bacigal, T. (eds.), *Proc. Uncertainty Modeling*, STU Bratislava, 2013, pp. 17–23

P81. Cena, A., Gagolewski, M., *OM3: Ordered maxitive, minitive, and modular aggregation operators – Part I: Axiomatic analysis under arity-dependence*, in: Bustince, H. et al. (eds.), *Aggregation Functions in Theory and in Practise*, Springer, 2013, pp. 93–103

P82. Cena, A., Gagolewski, M., *OM3: Ordered maxitive, minitive, and modular aggregation operators – Part II: A simulation study*, in: Bustince, H. et al. (eds.), *Aggregation Functions in Theory and in Practise*, Springer, 2013, pp. 105–115

P83. Gagolewski, M., *Statistical hypothesis test for the difference between Hirsch indices of two Pareto-distributed random samples*, in: Kruse, R. et al. (eds.), *Synergies of Soft Computing and Statistics for Intelligent Data Analysis*, Springer, 2013, pp. 359–367

P84. Gagolewski, M., *On the relation between effort-dominating and symmetric minitive aggregation operators*, in: Greco, S. et al. (eds.), *Advances in Computational Intelligence, Part III*, Springer, 2012, pp. 276–285

P85. Gagolewski, M., Grzegorzewski, P., *Axiomatic characterizations of (quasi-) L-statistics and S-statistics and the Producer Assessment Problem*, in: Galichet, S. et al. (eds.), *Proc. EUSFLAT/LFA'11*, Atlantis Press, 2011, pp. 53–58

P86. Rowiński, T., Gagolewski, M., *Internet a kryzys*, in: Jankowska, M., Starzomska, M. (eds.), *Kryzys: Pułapka czy szansa?*, WN Akapit, Warsaw, 2011, pp. 211–224

P87. Gagolewski, M., Grzegorzewski, P., *Arity-monotonic extended aggregation operators*, in: Hüllermeier, E. et al. (eds.), *Information Processing and Management of Uncertainty in Knowledge-Based Systems*, Springer, 2010, pp. 693–702

P88. Gagolewski, M., Grzegorzewski, P., *S-statistics and their basic properties*, in: Borgelt, C. et al. (eds.), *Combining Soft Computing and Statistical Methods in Data Analysis*, Springer, 2010, pp. 281–288

P89. Gagolewski, M., Grzegorzewski, P., *Metody i problemy naukometrii (Methods and problems of scientometrics)*, in: Rowiński, T., Tadeusiewicz, R. (eds.), *Psychologia i informatyka. Synergia i kontradykcje*, Wyd. UKSW, Warsaw, 2010, pp. 103–125

P90. Gagolewski, M., Grzegorzewski, P., *O pewnym uogólnieniu indeksu Hirscha*, in: Kawalec, P., Lipski, P. (eds.), *Kadry i infrastruktura nowoczesnej nauki: Teoria i praktyka, Proc. 1st Intl. Conf. Zarządzanie Nauką*, Wydawnictwo Lubelskiej Szkoły Biznesu, Lublin, 2009, pp. 15–29

P91. Gagolewski, M., Grzegorzewski, P., *Possible and necessary h-indices*, in: Carvalho, J. et al. (eds.), *Proc. IFSA/EUSFLAT'09*, IFSA, 2009, pp. 1691–1695

6.2.3 Edited Volumes (3)

P92. Halaš, R., Gagolewski, M., Mesiar, R. (eds.), *New Trends in Aggregation Theory*, Springer, 2019, 348 pp.

P93. Ferraro, M., Giordani, P., Vantaggi, B., Gagolewski, M., Gil, M., Grzegorzewski, P., Hryniewicz, O. (eds.), *Soft Methods for Data Science*, Springer, 2017, 535 pp.

P94. Grzegorzewski, P., Gagolewski, M., Hryniewicz, O., Gil, M. (eds.), *Strengthening Links Between Data Analysis and Soft Computing*, Springer, 2015, 294 pp.

6.2.4 Research Monographs and Textbooks (7)

P95. Gagolewski, M., *Deep R Programming*, Melbourne, v1.0.2 edition, 2026, URL: https://deepr.gagolewski.com/

P96. Gagolewski, M., *Minimalist Data Wrangling with Python*, Melbourne, v1.1.0 edition, 2026, URL: https://datawranglingpy.gagolewski.com/

P97. Gagolewski, M., *Algorytmy i postawy programowania w języku C++ (Introduction to Algorithms and Programming in C++)*, Melbourne, v1.2.0 edition, 2022, 209 pp., URL: https://github.com/gagolews/aipp

P98. Gagolewski, M., Bartoszuk, M., Cena, A., *Przetwarzanie i analiza danych w języku Python (Data Processing and Analysis in Python)*, Wydawnictwo Naukowe PWN, Warsaw, 2016, 369 pp., URL: https://github.com/gagolews/Analiza_danych_w_jezyku_Python

P99. Gagolewski, M., *Programowanie w języku R. Analiza danych, obliczenia, symulacje (R Programming. Data Analysis, Computing, Simulations)*, Wydawnictwo Naukowe PWN, Warsaw, 2nd edition, 2016, 550 pp., URL: https://github.com/gagolews/Programowanie_w_jezyku_R

P100. Gagolewski, M., *Data Fusion: Theory, Methods, and Applications*, Institute of Computer Science, Polish Academy of Sciences, Warsaw, 2015, 290 pp., URL: https://github.com/gagolews/datafusion

P101. Grzegorzewski, P., Gagolewski, M., Bobecka-Wesołowska, K., *Wnioskowanie statystyczne z wykorzystaniem środowiska R (Statistical Inference with R)*, Politechnika Warszawska, Warsaw, 2014, 183 pp.

6.3 Talks (Conferences, Seminars, etc.)

6.3.1 Invited Plenary Lectures and Tutorials

1. *Jak badać metody analizy skupień? (Methods for analysis of cluster analysis methods)*, XLIX Konferencja „Statystyka Matematyczna – Chęciny 2024", Chęciny, Poland, 09–12.12.2024 (In Polish)
2. *Clustering and aggregation*, 16th International Conference on Fuzzy Set Theory and Applications – FSTA 2022, Liptovský Ján, Slovakia, 04.02.2022 (online)
3. *Clustering on MSTs*, International Student Conference on Applied Mathematics and Informatics ISCAMI'18, Malenovice, Czechia, 10–13.05.2018
4. *Stochastic properties of and agent-based models for the Hirsch index and other discrete Sugeno integrals*, 14th International Conference on Fuzzy Set Theory and Applications – FSTA 2018, Liptovský Ján, Slovakia, 02.02.2018
5. *Aggregation of multidimensional data: A review*, 9th International Summer School on Aggregation Operators – AGOP 2017, Skövde, Sweden, 21.06.2017
6. *Penalty-based fusion of complex data, computational aspects, and applications*, International Symposium on Aggregation and Structures – ISAS 2016, University of Luxembourg, 06.07.2016

6.3.2 Other Invited

7. *R package* stringi, Text Analysis Developers' Workshop 2018, New York University, New York, NY, US, 20–21.04.2018
8. *Algorytmy analizy skupień oparte na MST*, Studencka konferencja zastosowań matematyki DwuMIan'18, Warsaw, Poland, 24.03.2018
9. *R package* stringi, Text Analysis R Developers' Workshop 2017, London School of Economics, London, England, 21–22.04.2017
10. Genie*: A new, fast, and outlier-resistant hierarchical clustering algorithm and its R interface*, European R Users Meeting, Poznań, Poland, 14.10.2016
11. *Can the scientific assessment process be fair?*, Workshop on Research Evaluation, Free University of Bozen-Bolzano, Italy, 10.05.2013

6.3.3 Seminars

12. *Aggregation of multidimensional data: A review*, School of Information Technology, Deakin University, Melbourne-Burwood, VIC, Australia, 21.07.2017
13. *Agregacja danych: Teoria, metody i zastosowania*, Wykład dla słuchaczy Studiów Doktoranckich IBS PAN, Warszawa, Poland, 05.03.2016
14. `^(R|ICU|i18n|regex)+$`, Seminarium Matematyczne Metody Informatyki, Instytut Matematyki, University of Silesia, Katowice, Poland, 20.04.2015
15. *Data aggregation from an algorithmic perspective*, IRAFM Seminar, University of Ostrava, Czechia, 04.06.2015
16. *Indeks Hirscha i okolice*, Seminarium CeON, ICM UW, Warsaw, Poland, 12.03.2014
17. *Scientific impact assessment – State of the art: Agregačné funkcie: teória a aplikácie* (*Aggregation functions: theory and applications*), Seminár z modelovania neurčitosti, Katedra matematiky a deskriptívnej geometrie, SvF STU, Bratislava, Slovakia, 17.04.2013

6.3.4 Conference Talks

18. *Yet another graphical aid to the interpretation and validation of cluster analysis*, 6th International Symposium on Fuzzy Sets, Katowice, 23–25.05.2025
19. *Penalty-based data aggregation in real normed vector spaces*, 10th International Summer School on Aggregation Operators (AGOP), Olomouc, Czechia, 01–04.07.2019
20. *Fitting symmetric fuzzy measures for discrete Sugeno integration*, 10th International Conference of EUSFLAT, Warsaw, Poland, 11–15.09.2017
21. *Binary aggregation functions in software plagiarism detection*, IEEE International Conference on Fuzzy Systems (IEEE FUZZ'17), Naples, Italy, 09–12.07.2017
22. *Binary aggregation functions in software plagiarism detection*, 3rd International Symposium on Fuzzy Sets and Uncertainty Modeling (ISFS 2017), Rzeszów, Poland, 19–20.05.2017
23. *Hierarchical clustering via penalty-based aggregation and the Genie approach*, 13th International Conference on Modeling Decisions for Artificial Intelligence (MDAI), Sant Julià de Lòria, Andorra, 20.09.2016
24. *Fitting aggregation functions to data: Part I – Linearization and regularization*, 16th International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems (IPMU), Eindhoven, The Netherlands, 23.06.2016
25. *Some issues in aggregation of multidimensional data*, 8th International Summer School on Aggregation Operators (AGOP), Katowice, Poland, 07.07.2015
26. *Normalized WD_pWAM and WD_pOWA spread measures*, International Conference of IFSA/EUSFLAT 2015, Gijon, Spain, 02.07.2015
27. *Sugeno integral-based confidence intervals for the theoretical h-index*, 7th International Conference on Soft Methods in Probability and Statistics (SMPS), Warsaw, Poland, 24.09.2014
28. *OM3: Ordered maxitive, minitive, and modular aggregation operators – Part I: Axiomatic analysis under arity-dependence*, 7th International Summer School on Aggregation Operators (AGOP), Pamplona, Spain, 16–19.07.2013

29. *Statistical hypothesis test for the difference between Hirsch indices of two Pareto-distributed random samples*, 6th International Conference on Soft Methods in Probability and Statistics (SMPS), Konstanz, Germany, 04–06.10.2012
30. *On the relation between effort-dominating and symmetric minitive aggregation operators*, 14th International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems (IPMU), Catania, Italy, 09–13.07.2012
31. *Porównanie wybranych estymatorów teoretycznego indeksu Hirscha*, XXXVII Konferencja Statystyka Matematyczna, Wisła, Poland, 05–09.12.2011
32. *Axiomatic characterizations of (quasi-) L-statistics and S-statistics and the Producer Assessment Problem*, 7th International Conference of EUSFLAT/LFA, Aix-Les-Bains, France, 18–22.07.2011
33. *Podstawowe właściwości S-statystyk*, XXXVI Konferencja Statystyka Matematyczna, Wisła, Poland, 06–10.12.2010
34. *S-Statistics and their basic properties*, 5th International Conference on Soft Methods in Probability and Statistics (SMPS), Oviedo, Spain, 28.09–01.10.2010
35. *Arity-monotonic extended aggregation operators*, 13th International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems (IPMU), Dortmund, Germany, 28.06–02.07.2010
36. *Uogólniony indeks Hirscha a dwupróbkowe testy dla rodziny rozkładów Pareto II rodzaju*, XXXV Konferencja Statystyka Matematyczna, Wisła, Poland, 07–11.12.2009
37. *O pewnym uogólnieniu indeksu Hirscha*, 1st International Conference on "Scientific Management", Lublin, Poland, 20–22.11.2009
38. *Possible and necessary h-indices*, 6th International Conference of IFSA/EUSFLAT, Lisbon, Portugal, 20–24.07.2009

6.4 Research Students

- I was the principal supervisor of the following PhD students:

  1. Jan Lasek (PhD – completed 2019; Warsaw University of Technology) – *New Data-Driven Rating Systems for Association Football*
  2. Maciej Bartoszuk (PhD cum laude – completed 2018; Warsaw University of Technology) – *A Source Code Similarity Assessment System for Functional Programming Languages Based on Machine Learning and Data Aggregation Methods* (in Polish)
  3. Anna Cena (PhD – completed 2018; Systems Research Institute, Polish Academy of Sciences) – *Adaptive Hierarchical Clustering Algorithms Based on Data Aggregation Methods* (in Polish)

- Current PhD students:

  1. Łukasz Brzozowski (PhD cand., 2022–; Warsaw University of Technology)

- I was the supervisor of the following MSc (by Research) students (Warsaw University of Technology):

  1. Dawid Stelmach (MSc – completed 2020) – *Video Anomaly Detection as a One-Class-Classification Problem* (in Polish)
  2. Maciej Kurek (MSc – completed 2020) – *Survival Analysis of the Time-to-Score in Sports* (in Polish)
  3. Piotr Wawrzyniak (MSc – completed 2019) – *Boxing Board Results Prediction Based on Neural Networks* (in Polish)
  4. Michał Hadryś (MSc – completed 2019) – *Comparison of models for match outcome prediction* (in Polish)
  5. Piotr Smuda (MSc – completed 2018) – *A Music Recommendation System* (in Polish)
  6. Mateusz Jabłoński (MSc – completed 2016) – *Dynamic Report Generation based on Jupyter Kernels* (in Polish)
  7. Natalia Potocka (MSc – completed 2016) – *Text Clustering Based on String Metrics* (in Polish)
  8. Piotr Frukacz (MSc – completed 2015) – *Mobile Salesman Assistant on Salesforce platform and Google API* (in Polish)
  9. Norbert Ryciak (MSc – completed 2015) – *A Text Topic Modelling-Based Recommender System Utilising the Latent Dirichlet Allocation Method* (in Polish)
  10. Emma Sanderson (MSc – completed 2015) – *New Methods for Calculating Optimal Safety Stocks at Procter&Gamble*
  11. Dawid Janocha (MSc – completed 2014) – *Continuous Integration in Software Engineering* (in Polish)

7 Reviewing and Other Academic Activities

7.1 Expert panels

- Area editor (aggregation functions and data science) in *Fuzzy Sets and Systems*, 2021–
- International expert panel member (Natural Sciences 2), Slovak Research and Development Agency (Agentúra na podporu výskumu a vývoja – APVV), 2024–
- Expert panel member, The Science Evaluation Committee, Ministry of Science and Higher Education, Poland, 2026 (Komisja Ewaluacji Nauki – Ewaluacja jednostek naukowych 2022–2025 – ocena ekspercka w kryterium K-I i K-II)
- Member of the Research Council (Computer Science and Telecommunication), Warsaw University of Technology, 2019–
- Member of the Scientific Council, Systems Research Institute, Polish Academy of Sciences, 2011–
- Member of the Faculty Council, Faculty of Mathematics and Information Science, Warsaw University of Technology, 2017–

7.2 Reviews of research project proposals

1. Australian Research Council (ARC) (3)
2. The National Fund for Scientific and Technological Development of Chile (Fondo Nacional de Desarrollo Científico y Tecnológico – FONDECYT) (1)
3. Slovak Research and Development Agency (Agentúra na podporu výskumu a vývoja – APVV) (11)

7.3 Scientific program committee member/chair

1. 22nd Conference on Complex Systems (CCS 2026), Binghamton, NY, USA
2. 14th Conference of the European Society for Fuzzy Logic and Technology (EUSFLAT 2025), Riga, Latvia
3. 21st Conference on Complex Systems (CCS 2025), Siena, Italy
4. 20th Conference on Complex Systems (CCS 2024), London, England
5. 13th Conference of the European Society for Fuzzy Logic and Technology (EUSFLAT/AGOP/FQAS 2023), Palma, Mallorca, Spain
6. 11th International Summer School on Aggregation Operators (AGOP 2021), Bratislava, Slovakia – Program Chair
7. 19th World Congress of the International Fuzzy Systems Association and 12th Conference of the European Society for Fuzzy Logic and Technology (IFSA/EUSFLAT 2021), Bratislava, Slovakia
8. 10th International Summer School on Aggregation Operators (AGOP 2019), Olomouc, Czechia
9. 11th Conference of the European Society for Fuzzy Logic and Technology (EUSFLAT 2019), Prague, Czechia
10. 2nd International Symposium on Aggregation and Structures (ISAS 2018), Valladolid, Spain
11. 3rd Conference on Information Technology, Systems Research and Computational Physics (ITSRCP'18), Cracow, Poland
12. 17th World Congress of International Fuzzy Systems Association and 9th International Conference on Soft Computing and Intelligent Systems (IFSA/SCIS 2017), Otsu, Japan
13. 1st International Symposium on Aggregation and Structures (ISAS 2016), Luxembourg
14. 16th World Congress of the International Fuzzy Systems Association and 9th Conference of the European Society for Fuzzy Logic and Technology (IFSA/EUSFLAT 2015), Gijon, Spain

7.4 Conference organisation

1. IEEE World Congress on Computational Intelligence (WCCI 2020), Glasgow (UK) – special session FUZZ-IEEE-6 *Aggregation Structures: New Trends and Applications*
2. 10th International Summer School on Aggregation Operators (AGOP 2019), Olomouc, Czechia – conference chair

3. 10th Conference of the European Society for Fuzzy Logic and Technology (EUSFLAT 2017), Warsaw, Poland – Stream on Data Analysis Coordinator and special session *Algorithms for Data Aggregation and Fusion*
4. 16th International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems (IPMU 2016), Eindhoven, The Netherlands – special session *Computational Aspects of Data Aggregation and Complex Data Fusion*
5. 8th International Conference Soft Methods in Probability and Statistics (SMPS 2016), Rome, Italy– organising committee member
6. 8th International Summer School on Aggregation Operators (AGOP 2015), Katowice, Poland– organising committee member
7. 7th International Conference Soft Methods in Probability and Statistics – SMPS 2014, Warsaw, Poland– organising committee member
8. 37th Conference *Statystyka Matematyczna – Wisła 2011*, Poland– organising committee member

7.5 Reviewer/Examiner of PhD theses

1. dr Grzegorz Moś, University of Silesia in Katowice, Poland, 2026
2. dr Luke William Yerbury, School of Information and Physical Sciences, The University of Newcastle, Australia, 2026
3. Hossein Yazdani, Faculty of Electronics, Wrocław University of Science and Technology, Poland, 2018 and 2020 (re-review)
4. dr Jana Borzová, Faculty of Science, P. J. Šafárik University in Košice, Slovakia, 2018

7.6 PhD/DSc committee member

1. Arkadiusz Sycz, Warsaw University of Technology, t.b.d.
2. Michał Ostapowicz, Warsaw University of Technology, t.b.d.
3. dr Anthony Nwachukwu, Warsaw University of Technology, 2026
4. dr Olga Kamińska, Systems Research Institute, Polish Academy of Sciences, 2025
5. dr Kacper Kania, Warsaw University of Technology, 2025
6. dr Weronika Hryniewska-Guzik, Warsaw University of Technology, 2025
7. dr Weronika Gutfeter, Warsaw University of Technology, 2023
8. dr Tomasz Rybotycki, Systems Research Institute, Polish Academy of Sciences, 2023
9. dr Grzegorz Gołaszewski, Systems Research Institute, Polish Academy of Sciences, 2022
10. dr hab. Barbara Pękala, Systems Research Institute, Polish Academy of Sciences, 2019

7.7 Publication reviews

Peer-reviewer for the following international journals (324 reviews written):

1. *Accountability in Research: Ethics, Integrity and Policy* (1)
2. *ACM Computing Surveys* (1)
3. *ACM Transactions on Knowledge Discovery from Data* (1)
4. *ACM Transactions on Mathematical Software* (5)
5. *Advances in Statistical Analysis (German Statistical Society)* (3)
6. *Afrika Matematika* (1)
7. *Annals of Operations Research* (2)
8. *Bulletin of the Polish Academy of Sciences Technical Sciences* (1)
9. *Computational and Applied Mathematics* (3)
10. *Computers and Operations Research* (2)
11. *Control and Cybernetics* (1)

12. *Data Mining and Knowledge Discovery* (4)
13. *Demonstratio Mathematica* (1)
14. *Engineering Applications of Artificial Intelligence* (1)
15. *European Journal of Operational Research* (20)
16. *Expert Systems with Applications* (1)
17. *Foundations of Computing and Decision Sciences* (1)
18. *Fundamenta Informaticae* (1)
19. *Fuzzy Optimization and Decision Making* (3)
20. *Fuzzy Sets and Systems* (39)
21. *Group Decision and Negotiation* (1)
22. *IEEE Access* (1)
23. *IEEE Transactions on Big Data* (1)
24. *IEEE Transactions on Emerging Topics in Computational Intelligence* (2)
25. *IEEE Transactions on Fuzzy Systems* (61)
26. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (1)
27. *Information Fusion* (9)
28. *Information Sciences* (47)
29. *Intelligent Systems with Applications* (2)
30. *International Journal of Applied Mathematics and Computer Science* (5)
31. *International Journal of Approximate Reasoning* (4)
32. *International Journal of Computational Intelligence Systems* (4)
33. *International Journal of Forecasting* (1)
34. *International Journal of Sports Science and Coaching* (5)
35. *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems* (5)
36. *International Transactions in Operational Research* (1)
37. *Journal of Applied Analysis* (1)
38. *Journal of Engineering Education* (1)
39. *Journal of Informetrics* (13)
40. *Journal of Intelligent and Fuzzy Systems* (3)
41. *Journal of Open Source Software* (3)
42. *Journal of Quantitative Analysis in Sports* (1)
43. *Journal of the Association for Information Science and Technology* (7)
44. *Knowledge and Information Systems* (1)
45. *Knowledge-Based Systems* (3)
46. *Mathematical Problems in Engineering* (1)
47. *Neurocomputing* (3)
48. *Operations Research and Decisions* (1)
49. *Pervasive and Mobile Computing* (1)
50. *Physica A: Statistical Mechanics and its Applications* (1)
51. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* (1)
52. *Quantitative Science Studies* (1)
53. *R Journal* (4)
54. *RUDN Journal of Mathematics, Information Sciences and Physics* (1)
55. *Scientific Reports* (1)
56. *Scientometrics* (26)
57. *Social Network Analysis and Mining* (1)

58. *Social Sciences and Humanities Open* (1)
59. *Soft Computing* (3)
60. *Statistical Modelling* (2)
61. *Wiley Interdisciplinary Reviews (WIREs) Data Mining and Knowledge Discovery* (1)

and international conferences (94 reviews written; IFSA/EUSFLAT 2009, IPMU 2010, IPMU 2012, SMPS 2014, EUSFLAT 2015, IPMU 2016, ISAS 2016, SMPS 2016, EUSFLAT 2017, IFSA/SCIS 2017, EUSFLAT 2019, FUZZ-IEEE 2020, IPMU 2020, AGOP 2021, FUZZ-IEEE 2021, IFSA/EUSFLAT 2021, FUZZ-IEEE 2022, IEEE-SAIC 2022, EUSFLAT 2023, FUZZ-IEEE 2023, Complex Systems 2024, IPMU 2024, Complex Systems 2025, EUSFLAT 2025, Complex Systems 2026, PP-RAI 2026, SMPS 2026)

8 Teaching-Related Activities

8.1 Faculty of Mathematics and Information Science, Warsaw University of Technology (Poland)

- (Founding) Vice-chair of the Teaching and Learning Committee for the BSc and MSc degrees in Data Science (2016–2019, 2024–); key responsibilities: developing and supervising a new degree in Data Science
- Supervisor of the Data Science studies (a.k.a. Dean's Proxy for Data Science Studies) (2018–2019, 2024–); key responsibilities:
  - managing, assessing, and synchronising all data science subjects
  - handling undergraduate and graduate students' admissions
  - coordinating students' transfers, providing advice regarding degree requirements and exchange programs (such as within the Erasmus framework)
  - lecturer/teacher allocation
  - counselling students with regards to their academic goals and how to meet them
- Member of the Teaching and Learning Committee for the BSc degree in Mathematics and Data Analysis (2019–)
- Initiator, supervisor, and mentor of the *Data Science* Student Club (2014–2019)
- Courses (units, subjects) taught:
  - Implementing software libraries and packages for data science
    New Unit Developer, Chair, and Lecturer (1 edition), 2026–
  - Computational Data Science aka Advanced Algorithms and Data Structures for Data Science
    New Unit Developer, Chair, and Lecturer (2 editions), 2024–
  - Research Thesis Seminar; BSc in Data Science
    Coordinator (2 editions), 2024–
  - Structured Data Processing
    New Unit Developer, Chair, and Lecturer (4 editions), 2018–2019, 2024–2025
  - Introduction to Programming and Data Processing
    New Unit Developer, Chair, and Lecturer (2 editions), 2017–2019
  - Data Processing in R and Python
    New Unit Developer, Chair, and Lecturer (3 editions), 2016–2019
  - Data Processing and Analysis in Python
    New Unit Developer, Chair, and Lecturer (4 editions), 2015–2019
  - Programming and Data Analysis in R
    New Unit Developer, Chair, and Lecturer (7 editions), 2012–2019

- Algorithms and Introduction to Programming
  New Unit Developer, Chair, and Lecturer (6 editions), 2010–2016
- Advanced R Programming
  New Unit Developer, Chair, and Lecturer (1 edition), 2013–2014
- Mathematical Statistics I
  Tutor (4 editions), 2009–2012
- Computer statistics
  Tutor (5 editions), 2008–2013
- Programming in x86 Assembler
  Tutor (2 editions), 2010–2011
- Algorithms and Data Structures II
  Tutor (4 editions), 2008–2011
- Object-oriented Programming in C++
  Tutor (5 editions), 2007–2011

8.2 Deakin University, School of IT (Australia)

- Deputy Course Director for BSc in Data Science, 2022–2024
- Course Leadership Team Member for BSc in Artificial Intelligence, 2020–2023
- Course Leadership Team Member for MSc of Data Science, 2023–2024
- Academic mentor of 10 industry capstone projects at Deakin University – providing mentoring and assessment of students, 2020–2021
- Courses (units, subjects) taught:
  - SIT220/731 – Data Wrangling
    New Unit Developer, Chair, and Lecturer in 2022.T1, 2022.T3, 2023.T1, and 2023.T3
  - SIG731 – Data Wrangling (Master of Data Science (Global, online); via "Great Learning" in India)
    Unit Chair and Guest Lecturer in 2022.T3 and 2023.T3
  - SIT114 – Introduction to Artificial Intelligence
    New Unit Developer, Chair, and Lecturer in 2020.T1, 2021.T1, and 2022.T1
  - SIT752 – Introduction to IT Professional Practice
    Unit Chair in 2019.T3 and 2020.T1
  - SIT172 – Programming for Engineers
    Unit Campus Coordinator and Co-lecturer in 2020.T2 and 2021.T2

8.3 Other

Courses/units taught:

- Data Science Retreat, Berlin
  - NumPy, Pandas, TensorFlow, Advanced Python, Data Structures and Algorithms for Data Science, Introduction to R, Advanced R, Rcpp, Speeding up R and Python, String Processing, Good Development Practices in R
    Guest Lecturer (19 student batches), 2014–2019
- University of Silesia in Katowice
  - Introduction to Data Science in Python (online)
    Guest Lecturer, 2022

- Centre for Advanced Studies, Warsaw University of Technology
  - Python for Data Processing and Analysis
    Unit Developer and Chair in 2018
- Institute of Computer Science, Polish Academy of Sciences
  - Advanced Data Analysis Software Development in R (online)
    Unit Developer and Chair; three editions, 2014–2015
- Warsaw School of Information Technology
  - Statistical Decision Support Methods
    Tutor; three semesters, 2009–2011
  - Probability and Statistics
    Tutor; three semesters, 2008–2010

Other:

- Amongst top 3% StackOverflow users (https://stackoverflow.com/users/3309529/gagolews)
- StackOverflow *Academic Research Partnership Program* – Supervisor of a research task related to quantitative determinants of the popularity of online content, 2019
- Google *Summer of Code 2016* – Mentor of the *RE2 Regular Expressions in R* project (Student: Qin Wenfeng), 2016

8.4 Diploma theses

- Principal supervisor of 21 B.Sc. students in Mathematics, Computer Science, and Data Science
- Principal supervisor of 11 MSc (by research) students in Mathematics, Computer, and Data Science

9 Open-Source Software Development

Author and maintainer of free (libre) and open-source source software (see my GitHub profile at https://github.com/gagolews/), e.g.,:

O1. *stringi* (https://stringi.gagolewski.com/) – text/natural language processing; one of the most often downloaded R packages (over 100,000,000 downloads)
O2. *genieclust* (https://genieclust.gagolewski.com/) – Python and R implementation of my fast and robust *Genie* hierarchical clustering algorithm
O3. *clustering-benchmarks* (https://clustering-benchmarks.gagolewski.com/) – A framework for benchmarking clustering algorithms (including a package for Python)
O4. *deadwood* (https://deadwood.gagolewski.com/) – Outlier detection via trimming of mutual reachability minimum spanning trees in R and Python
O5. *lumbermark* (https://lumbermark.gagolewski.com/) – Clustering via chopping up mutual reachability minimum spanning trees in R and Python
O6. *quitefastmst* (https://quitefastmst.gagolewski.com/) – Euclidean and mutual reachability minimum spanning trees in R and Python
O7. *stringx* (https://stringx.gagolewski.com/) – drop-in replacements for base R string functions powered by *stringi*
O8. *realtest* (https://realtest.gagolewski.com/) – a framework for unit testing for realistic minimalists, where we distinguish between expected, acceptable, current, fallback, ideal, or regressive behaviour; it can also be used for monitoring other software projects for changes

O9. *FuzzyNumbers* (http://cran.r-project.org/package=FuzzyNumbers) – tools to deal with fuzzy numbers in R

O10. *TurtleGraphics* (http://cran.r-project.org/package=TurtleGraphics) – learn R programming while having a jolly time!

Marek Gagolewski
19 October 2026